UNIT 6 · STANDARD 6.EE.3

Properties of Operations

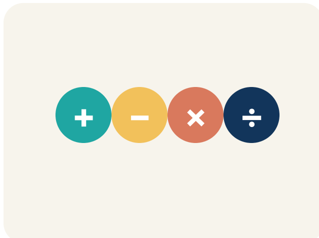
Lesson 6-4

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

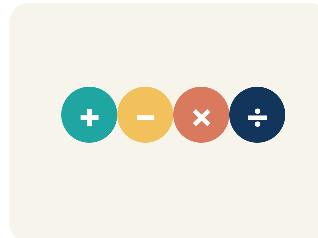
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$3 + 7 = 7 + 3 = 10$ both ways; $4 \times 5 = 5 \times 4 = 20$ both ways — order does not matter

Commutative Property

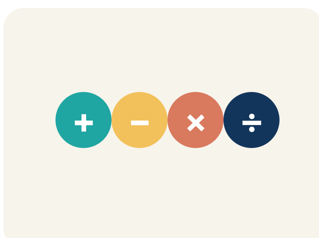
You can change the order and get the same answer.



$(2 + 3) + 4 = 5 + 4 = 9$ and $2 + (3 + 4) = 2 + 7 = 9$ — same answer, different grouping

Associative Property

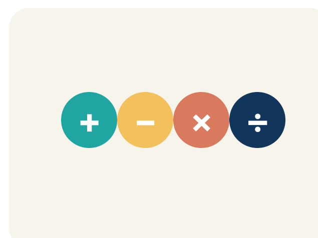
You can change the grouping and get the same answer.



$9 + 0 = 9$ (zero is the identity for addition); $6 \times 1 = 6$ (one is the identity for multiplication)

Identity Property

Adding 0 or multiplying by 1 keeps the same value.



Commutative works for $+$ and $\times$, but NOT for $-$ or $\div$ (since $5 - 3 = 2$ but $3 - 5 = -2$)

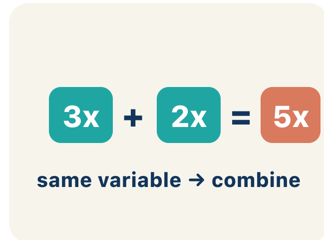
Property

A rule that is always true in math.

In 3x, the 3 is the coefficient — it tells you to multiply x by 3

Coefficient

The number in front of a letter, like the 3 in 3x.



4x + 2x = 6x (like terms, both have x); 4x + 2y stays as 4x + 2y (unlike terms)

Like terms

Terms with the same letter, like 2x and 5x.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can use the commutative, associative, and identity properties to rewrite expressions.

1️⃣ Notice

You're a band director arranging musicians on stage.

2️⃣ Key idea

I can use the commutative, associative, and identity properties to rewrite expressions.

3️⃣ Words I'll use

- **Commutative Property**
- **Associative Property**
- **Identity Property**
- **Property**
- **Coefficient**
- **Like terms**

🤔 Think About It

- Does the order you add the musicians change the total?
- What if you group different sections together first — does the total change?
- What happens when you add 0 musicians to a group?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Which property is shown? $5 + 13 = 13 + 5$

A. Commutative Property

B. Associative Property

C. Identity Property

D. Distributive Property

Step 1 The order of the addends changed, so this is the Commutative Property of Addition.

✅ **Answer:** A. Commutative Property

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which property is shown? $(6 \times 3) \times 2 = 6 \times (3 \times 2)$

A. Associative Property

B. Commutative Property

C. Identity Property

D. Distributive Property

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which property is shown? $47 + 0 = 47$

A. Identity Property

B. Commutative Property

C. Associative Property

D. Distributive Property

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The band has 3 guitarists, 5 drummers, and 2 keyboard players. Whether you seat guitarists or drummers first, you still have 10 musicians. Which property explains this, and why does it work?

Level 1 support

Start here: The commutative property lets you change the order of a sum without changing the total.

💡 **Need a hint?**

Sentence starters (optional)

Changing the order still gives 10 because of the ____ property.

Cambiar el orden todavía da 10 por la propiedad ____.

It works because order does not change ____.

Funciona porque el orden no cambia ____.

Word bank: Commutative Property | Associative Property | Identity Property | Property

Listen for: *Students name the commutative property and explain that 3 + 5 + 2 gives 10 in any order because reordering addends does not change the sum.*

Level 2

Would this same reordering trick work if you were subtracting musicians who left the stage? Justify why the commutative property fails for subtraction.

- Reordering ____ work for subtraction because ____.
- For example, $5 - 3 =$ ____ but $3 - 5 =$ ____, so ____.

EXPLORE

How can you tell the commutative property from the associative property? Use 6 + 9 = 9 + 6 and (3 + 5) + 2 = 3 + (5 + 2) to explain.

Level 1 support

Start here: Commutative changes the order; associative changes the grouping.

💡 **Need a hint?**

Sentence starters (optional)

The ___ property changes the order, like 6 + 9 = 9 + 6.

La propiedad ___ cambia el orden, como 6 + 9 = 9 + 6.

The ___ property changes the grouping because ___.

La propiedad ___ cambia la agrupación porque ___.

Word bank: Commutative Property | Associative Property | Identity Property | Property

Listen for: *A strong answer connects commutative to changed order and associative to changed parentheses (grouping), noting both keep the same value.*

Level 2

Explain why the identity property (like 14 + 0 = 14) is a different idea from both. Generalize what each of the three properties keeps the same.

- The identity property is different because it ___ rather than reorder or regroup.
- All three keep the value the same, but they change ___, ___, or ___.

CONNECT

To find $4 \times 17 \times 25$ mentally, the engineer rearranges to $4 \times 25 \times 17 = 100 \times 17$. Which properties did she use, and why does rearranging help?

Level 1 support

Start here: Use commutative and associative properties to make a friendly factor of 100.

💡 **Need a hint?**

Sentence starters (optional)

She used the ____ property to change the order and the ____ property to regroup.

Usó la propiedad ____ para cambiar el orden y la propiedad ____ para reagrupar.

This helps because $4 \times 25 =$ ____, which is ____.

Esto ayuda porque $4 \times 25 =$ ____, que es ____.

Word bank: Commutative Property | Associative Property | Property | Identity Property

Listen for: *Students name both the commutative and associative properties, compute $4 \times 25 = 100$, and explain pairing to a friendly number makes the mental math easier ($100 \times 17 = 1700$).*

Level 2

Could you use the same strategy to compute $2 \times 13 \times 50$ quickly? Justify which numbers you would pair and why.

- I would pair ____ and ____ because their product is ____.
- Rearranging works here because the properties let me ____ without changing the answer.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words commutative property, associative property, and identity property.

1. Kernel Sentence

subject + verb

Model: Associative Property is you can change the grouping and get the same answer.
Propiedad asociativa es puedes cambiar la agrupación y obtener la misma respuesta.

Write a kernel sentence about associative property. Use a subject and a verb.
Escribe una oración base sobre propiedad asociativa. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Associative Property matters in math
Propiedad asociativa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Associative Property matters in math because ____.
Propiedad asociativa importa en matemáticas porque ____.

___

but
pero

Associative Property matters in math, but ____.
Propiedad asociativa importa en matemáticas, pero ____.

___

so
entonces

Associative Property matters in math, so ____.
Propiedad asociativa importa en matemáticas, entonces ____.

___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about associative property.
Di un hecho verdadero sobre associative property.

Associative property ____.

Question
Pregunta

Ask a question about associative property.
Haz una pregunta sobre associative property.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about associative property.
Muestra entusiasmo sobre associative property.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with associative property.
Dile a un compañero qué hacer con associative property.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

Changing the order still gives 10 because of the ____ property.
Cambiar el orden todavía da 10 por la propiedad ____.

It works because order does not change ____.
Funciona porque el orden no cambia ____.

The ____ property changes the order, like 6 + 9 = 9 + 6.
La propiedad ____ cambia el orden, como 6 + 9 = 9 + 6.

Try It

Solve on your own. Check the answer key when you are done.

1. What number completes the Identity Property: 9 × ____ = 9?

A. 1

B. 0

C. 9

D. 81

Show your work:

2. Which equation shows the Distributive Property?

A. $3(x + 2) = 3x + 6$

B. $3 + x = x + 3$

C. $(3x)y = 3(xy)$

D. $x + 0 = x$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Explain why the commutative property works for addition and multiplication but NOT for subtraction and division. Give a specific number example for each operation to prove your point.

Sentence starter: Addition: ____ + ____ = ____ + ____ = ____ (works). Subtraction: ____ − ____ ≠ ____ − ____ because ____. Multiplication: ____ × ____ = ____ × ____ = ____ (works). Division: ____ ÷ ____ ≠ ____ ÷ ____ because ____.

Show your work:

Reflect — Exit Ticket

Which property is shown? (8 + 5) + 2 = 8 + (5 + 2)

A. Associative Property

B. Commutative Property

C. Identity Property

D. Distributive Property

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. Associative Property *— The grouping (parentheses) changed but the order stayed the same, so this is the Associative Property of Multiplication.*
2. **You Try (notes):** A. Identity Property *— Adding 0 does not change the value, so this is the Identity Property of Addition.*
3. **Try It 1:** A. 1 *— Multiplying by 1 keeps the value the same.*
4. **Try It 2:** A. $3(x + 2) = 3x + 6$ *— Distributing 3 over $(x + 2)$ gives $3x + 6$.*
5. **Exit Ticket:** A. Associative Property *— The grouping (parentheses) changed but the numbers stayed in the same order, so this is the Associative Property of Addition.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Associative Property is you can change the grouping and get the same answer.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).